Enterprise RAG Platform: Product Requirements Document (PRD)

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1. Executive Summary

1.1 Project Vision

The enterprise RAG (Retrieval-Augmented Generation) platform is designed to be the definitive, secure, and scalable AI-powered knowledge retrieval layer for modern organizations. It bridges the gap between vast, fragmented internal knowledge repositories and the intuitive, conversational interface of Large Language Models (LLMs). By grounding LLM responses in verified, authorized enterprise data, the platform eliminates hallucinations while preserving data security and compliance [1].

1.2 Business Problem

Large enterprises possess millions of internal documents distributed across multiple repositories including SharePoint, Confluence, internal wikis, PDFs, and policy documents. Employees struggle to efficiently locate accurate information due to fragmented knowledge, outdated search capabilities, and information overload. Traditional keyword search often returns hundreds of irrelevant documents while failing to understand semantic meaning. While LLMs provide a superior user experience, they cannot natively access private enterprise knowledge and introduce significant security and compliance challenges [2].

1.3 Target Customers

The platform targets enterprise organizations with strict security, governance, and compliance requirements. Primary verticals include financial services, healthcare, government, telecommunications, manufacturing, insurance, and energy sectors. These organizations typically manage 10,000 to 100,000+ employees and require SOC 2 Type II, ISO 27001, GDPR, and HIPAA compliance.

1.4 Success Criteria

The following measurable criteria define platform success within the first 12 months of launch:

Metric	Target	Measurement Method
Platform availability	99.9% uptime	Cloud provider SLA + internal monitoring
Time-to-insight reduction	50% decrease	User surveys + A/B testing
Hallucination rate	< 2% for grounded queries	Automated evaluation pipeline
Enterprise customers onboarded	50 organizations	CRM tracking
Mean time to first value	< 7 days from signup	Analytics pipeline
Grounding accuracy	> 98%	Citation validation pipeline

1.5 Expected Business Impact

The platform is expected to increase employee productivity by providing instant access to accurate internal knowledge, reduce operational costs associated with internal support and knowledge discovery, enhance compliance and security through strict access controls and audit logging, and accelerate decision-making across departments.

1.6 Value Proposition

The platform provides a secure, multi-tenant, and highly scalable AI interface that allows employees to converse with their organization's proprietary knowledge. It guarantees complete tenant isolation, enforces strict access controls during retrieval, and provides fully cited, hallucination-free responses. Unlike generic AI tools, this platform treats the LLM as an untrusted component within a defense-in-depth security architecture, ensuring that every retrieval and generation decision respects organizational security policies.

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2. Product Goals

2.1 Primary Goals

The platform has five primary goals that define its core value. First, it must enable intelligent enterprise search by allowing employees to query internal knowledge bases using natural language rather than exact keywords. Second, it must ensure secure AI-powered answers by grounding every response exclusively in authorized enterprise knowledge. Third, it must operate as a multi-tenant SaaS architecture that serves multiple organizations simultaneously with complete data and security isolation. Fourth, it must implement enterprise-grade security including layered defenses, identity-aware access control, prompt injection mitigation, and end-to-end encryption. Fifth, it must support regulatory compliance including GDPR, EU data residency, and enterprise auditability requirements.

2.2 Secondary Goals

Secondary goals provide extensibility and operational capabilities. The platform should provide integration capabilities with popular enterprise document repositories, offer comprehensive analytics and usage metrics for platform administrators, and support advanced search features including hybrid search and metadata filtering.

2.3 Non-Goals

The following items are explicitly excluded from scope to maintain focus on the core RAG platform value proposition: LLM training or fine-tuning of foundation models, consumer-facing chatbot functionality, internet-wide public search, autonomous AI agents capable of executing privileged business actions (e.g., executing financial transactions), and cross-tenant knowledge sharing.

2.4 Out-of-Scope Items

Hardware procurement and physical data center management, legal consultation regarding specific regional compliance laws, on-premise deployment (initial release is cloud-native only), and custom model training services are out of scope.

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3. Target Users

The platform serves a diverse range of users within enterprise organizations. Each user group interacts with the platform differently and has distinct requirements.

User Group	Primary Interaction	Key Requirements
Enterprise Employees	Query knowledge, read answers	Natural language interface, citations

Knowledge Workers	Research, synthesize	Advanced search, multi-document retrieval
Engineers	Query technical docs, codebases	Code-aware chunking, structured outputs
Support Teams	Resolve customer queries	Shared knowledge bases, chat templates
HR & Finance	Access policies, procedures	Document versioning, compliance
Legal & Compliance	Review regulatory docs	Audit trails, data residency
Executives	Synthesize reports	High-level summaries, dashboards
IT Administrators	Configure platform	Admin dashboard, user management
Organization Owners	Oversee adoption, ROI	Analytics, billing, usage reports

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4. User Personas

4.1 The Employee (Sarah, Age 32, Marketing Manager)

Sarah manages a team of 15 and regularly needs to reference company policies, brand guidelines, and competitive intelligence. She is frustrated by navigating complex folder structures on SharePoint and spends 30 minutes per day searching for information that should take seconds to find. Her goal is to quickly find accurate, current answers to specific questions using natural language. In a typical usage scenario, Sarah asks the platform, "What is the current per diem limit for international travel to Japan?" and receives an immediate, cited answer based on the latest HR policy document, with a link to the full source.

4.2 The Knowledge Administrator (David, Age 45, IT Manager)

David is responsible for maintaining the organization's knowledge base across 12 different departments. He is overwhelmed by the volume of unstructured data and struggles to ensure users access the correct versions of documents. His goal is to efficiently ingest, index, and manage organizational knowledge while ensuring data is secure and up-to-date. In a typical scenario, David uploads a new 500-page compliance manual, configures indexing policies, assigns metadata, and sets access controls so only authorized personnel in the legal department can query the document.

4.3 The Security Architect (Priya, Age 38, CISO)

Priya is responsible for the organization's overall security posture. She is concerned about data leakage, prompt injection attacks, and unauthorized access to sensitive documents. Her goal is to ensure strict tenant isolation, enforce role-based access control, and mitigate AI security risks. In a typical scenario, Priya reviews the platform's audit logs to verify that all retrieval operations correctly enforced user permissions and that no prompt injection attempts were successful. She also configures data residency policies to ensure EU employee data never leaves the European region.

4.4 The Software Engineer (James, Age 29, Backend Developer)

James needs to query internal API documentation, architecture diagrams, and runbooks. He wants to use the platform's REST API to build a custom integration that surfaces relevant documentation within his team's IDE. His goal is to get accurate, structured answers about internal systems. In a typical scenario, James asks the platform to generate an API integration script using the developer SDK, and the platform provides code examples grounded in the actual internal API documentation.

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5. User Stories

The following user stories are organized by functional domain and prioritized using the MoSCoW method. Each story includes a unique identifier for traceability.

5.1 Authentication (PRD-US-AUTH)

ID	Story	Priority
PRD-US-AUTH-001	As a user, I want to log in using my corporate SSO credentials (SSO/OIDC/SAML), so that I don't need to remember a separate password.	Must
PRD-US-AUTH-002	As a user, I want to be prompted for MFA when my security policy requires it, so that my account remains secure.	Must

PRD-US-AUTH-003	As a user, I want to be automatically logged out after a period of inactivity, so that my session is secure.	Should
PRD-US-AUTH-004	As a user, I want to see which organizations and workspaces I have access to after login, so that I can navigate to the right context.	Must
PRD-US-AUTH-005	As a user, I want to manage my session across multiple devices, so that I can access the platform from my laptop and phone.	Should

5.2 Organization Management (PRD-US-ORG)

ID	Story	Priority
PRD-US-ORG-001	As an organization owner, I want to create a new organization, so that I can manage my company's data separately.	Must
PRD-US-ORG-002	As an owner, I want to invite users via email or SSO provisioning, so that they can access the platform.	Must
PRD-US-ORG-003	As an owner, I want to configure organization-wide security policies, so that they apply to all workspaces.	Must
PRD-US-ORG-004	As an owner, I want to delete an organization after archiving all data, so that I can clean up unused accounts.	Could

PRD-US-ORG-005	As an owner, I want to transfer organization ownership to another user, so that continuity is maintained.	Should

5.3 Workspace Management (PRD-US-WS)

ID	Story	Priority
PRD-US-WS-001	As an admin, I want to create multiple workspaces (e.g., Engineering, HR, Legal), so that data is logically separated.	Must
PRD-US-WS-002	As a user, I want to switch between different workspaces from a single interface, so that I can query different knowledge bases.	Must
PRD-US-WS-003	As an admin, I want to assign specific users to specific workspaces, so that access is restricted appropriately.	Must
PRD-US-WS-004	As an admin, I want to configure workspace-specific settings (chunking, embeddings, LLM), so that each workspace is optimized.	Could
PRD-US-WS-005	As an admin, I want to archive a workspace without deleting its data, so that I can preserve historical knowledge.	Should

5.4 Document Management (PRD-US-DOC)

ID	Story	Priority

PRD-US-DOC-001	As a user, I want to upload a single document (PDF, DOCX, etc.), so that I can add it to the knowledge base.	Must
PRD-US-DOC-002	As an admin, I want to bulk upload multiple documents via zip or folder, so that I can quickly ingest a large dataset.	Must
PRD-US-DOC-003	As a user, I want to delete a document, so that outdated information is removed from search results.	Must
PRD-US-DOC-004	As a user, I want to see the upload status and indexing progress in real-time, so that I know when a document is ready.	Should
PRD-US-DOC-005	As a user, I want to view document metadata (size, type, upload date, owner), so that I can manage my documents.	Should
PRD-US-DOC-006	As a user, I want to preview a document before it is indexed, so that I can verify its content.	Should
PRD-US-DOC-007	As an admin, I want to set document-level access controls (ACLs), so that only authorized users can retrieve the document.	Must
PRD-US-DOC-008	As a user, I want to re-upload a newer version of a document, so that the knowledge base stays current.	Should
PRD-US-DOC-009	As an admin, I want to configure retention policies for documents, so that expired documents are automatically removed.	Could

5.5 Folder Management (PRD-US-FLD)

ID	Story	Priority
PRD-US-FLD-001	As a user, I want to create folders to organize documents, so that the knowledge base is structured.	Should
PRD-US-FLD-002	As a user, I want to move documents between folders, so that I can correct organization errors.	Should
PRD-US-FLD-003	As a user, I want to apply folder-level permissions, so that access is inherited by all documents within.	Could

5.6 Collections (PRD-US-COL)

ID	Story	Priority
PRD-US-COL-001	As a user, I want to create a collection of specific documents across folders, so that I can query a focused subset.	Should
PRD-US-COL-002	As an admin, I want to set permissions on a collection, so that only authorized users can query it.	Must
PRD-US-COL-003	As a user, I want to share a collection with specific colleagues, so that we can collaborate on a topic.	Could

5.7 Knowledge Bases (PRD-US-KB)

ID	Story	Priority

PRD-US-KB-001	As an admin, I want to create a knowledge base from one or more collections, so that I can define a queryable scope.	Must
PRD-US-KB-002	As an admin, I want to configure the embedding model for a knowledge base, so that search quality is optimized.	Could
PRD-US-KB-003	As an admin, I want to set the chunking strategy (fixed-size, semantic, hierarchical), so that context retrieval is accurate.	Could
PRD-US-KB-004	As an admin, I want to view knowledge base statistics (document count, chunk count, index size), so that I can monitor health.	Should

5.8 Search (PRD-US-SRC)

ID	Story	Priority
PRD-US-SRC-001	As a user, I want to search for documents using natural language, so that I can find relevant information quickly.	Must
PRD-US-SRC-002	As a user, I want to use filters (date, author, type, source system) in my search, so that I can narrow down results.	Should
PRD-US-SRC-003	As a user, I want to see a preview of each document in the search results, so that I can verify its relevance.	Should
PRD-US-SRC-004	As a user, I want to save a search query for future reuse, so that I can quickly access frequently used searches.	Could
PRD-US-SRC-005	As a user, I want to bookmark a specific search result, so that I	Could

	can return to it later.	

5.9 Chat (PRD-US-CHT)

ID	Story	Priority
PRD-US-CHT-001	As a user, I want to ask a question to the AI chatbot and receive a synthesized answer, so that I can get information quickly.	Must
PRD-US-CHT-002	As a user, I want to see the response stream in real-time, so that I don't have to wait for the entire answer.	Must
PRD-US-CHT-003	As a user, I want to stop the generation of a response mid-stream, so that I can change my query if needed.	Should
PRD-US-CHT-004	As a user, I want to regenerate the response to my question, so that I can get a different formulation.	Should
PRD-US-CHT-005	As a user, I want to continue the conversation from where it left off, so that I can build on previous context.	Must
PRD-US-CHT-006	As a user, I want to provide feedback (thumbs up/down) on a response, so that the system can improve.	Should

5.10 Conversation History (PRD-US-HIS)

ID	Story	Priority
PRD-US-HIS-001	As a user, I want to view my past conversations in a sidebar,	Must

	so that I can reference previous information.	
PRD-US-HIS-002	As a user, I want to delete a past conversation, so that I can manage my history.	Should
PRD-US-HIS-003	As a user, I want to pin important conversations to the top of my history, so that I can find them quickly.	Could
PRD-US-HIS-004	As a user, I want to rename a conversation for easier identification, so that I can organize my history.	Could

5.11 Citations (PRD-US-CIT)

ID	Story	Priority
PRD-US-CIT-001	As a user, I want to see inline citations (numbered references) in the AI's response, so that I can verify the source.	Must
PRD-US-CIT-002	As a user, I want to click on a citation and see the exact snippet from the source document, so that I can read the context.	Must
PRD-US-CIT-003	As a user, I want to open the full source document from a citation, so that I can read the complete context.	Should
PRD-US-CIT-004	As a user, I want to see a confidence score for each citation, so that I can assess the reliability of the answer.	Could

5.12 Prompt Library (PRD-US-PRO)

ID	Story	Priority
PRD-US-PRO-001	As a user, I want to save a prompt as a reusable template, so that I can use it again later.	Could
PRD-US-PRO-002	As an admin, I want to create a shared prompt library available to all users in the organization, so that queries are standardized.	Could
PRD-US-PRO-003	As a user, I want to browse and search the prompt library, so that I can find relevant templates.	Could

5.13 Agents (PRD-US-AGT)

ID	Story	Priority
PRD-US-AGT-001	As a developer, I want to define an agent workflow that calls the RAG platform and external tools, so that I can automate multi-step tasks.	Won't (Phase 3)
PRD-US-AGT-002	As an admin, I want to configure which tools an agent can use, so that I can restrict its actions.	Won't (Phase 3)
PRD-US-AGT-003	As a user, I want to trigger an agent workflow from the chat interface, so that I can get automated multi-step answers.	Won't (Phase 3)

5.14 Administration (PRD-US-ADM)

ID	Story	Priority
PRD-US-ADM-001	As an admin, I want to view a dashboard showing system	Should

	health metrics, so that I can ensure the platform is running smoothly.	
PRD-US-ADM-002	As an admin, I want to configure feature flags, so that I can roll out new features gradually.	Could
PRD-US-ADM-003	As an admin, I want to manage system-wide settings (default LLM, chunk size, embedding model), so that I can optimize performance.	Could
PRD-US-ADM-004	As an admin, I want to view and manage API keys, so that I can control programmatic access.	Must
PRD-US-ADM-005	As an admin, I want to configure rate limits per user or API key, so that I can prevent abuse.	Should

5.15 Billing (PRD-US-BIL)

ID	Story	Priority
PRD-US-BIL-001	As an owner, I want to view my current usage and billing status, so that I can manage costs.	Must
PRD-US-BIL-002	As an owner, I want to upgrade or downgrade my subscription tier, so that I can adjust to my needs.	Must
PRD-US-BIL-003	As an owner, I want to receive invoice notifications, so that I can process payments.	Should

PRD-US-BIL-004	As an owner, I want to view a breakdown of usage by workspace, so that I can allocate costs to departments.	Could

5.16 Audit Logs (PRD-US-AUD)

ID	Story	Priority
PRD-US-AUD-001	As a security admin, I want to view an immutable log of all user actions (queries, uploads, permission changes), so that I can ensure compliance.	Must
PRD-US-AUD-002	As a security admin, I want to export audit logs in CSV or JSON format, so that I can perform external audits.	Must
PRD-US-AUD-003	As a security admin, I want to configure log retention periods, so that I comply with data retention policies.	Should
PRD-US-AUD-004	As a security admin, I want to receive alerts for suspicious activity (e.g., bulk downloads, unauthorized access), so that I can respond quickly.	Should

5.17 Analytics (PRD-US-ANA)

ID	Story	Priority
PRD-US-ANA-001	As an admin, I want to see which documents are queried most frequently, so that I can prioritize updating them.	Should

PRD-US-ANA-002	As an admin, I want to see search failure rates and zero-result queries, so that I can improve the knowledge base.	Should
PRD-US-ANA-003	As an admin, I want to view user adoption metrics (DAU, WAU, MAU), so that I can track platform engagement.	Should
PRD-US-ANA-004	As an owner, I want to see a dashboard of response quality metrics, so that I can assess the platform's value.	Could

5.18 Settings (PRD-US-SET)

ID	Story	Priority
PRD-US-SET-001	As a user, I want to change my notification preferences, so that I only receive relevant alerts.	Should
PRD-US-SET-002	As an admin, I want to configure the default LLM provider for the organization, so that I can control costs.	Could
PRD-US-SET-003	As a user, I want to set my preferred language for the interface, so that I can use the platform in my native language.	Should

5.19 API Usage (PRD-US-API)

ID	Story	Priority
PRD-US-API-001	As a developer, I want to view my API usage metrics (calls, tokens, latency), so that I can monitor my integration's load.	Should

PRD-US-API-002	As a developer, I want to set usage alerts, so that I am notified before hitting rate limits.	Could

5.20 Developer Portal (PRD-US-DEV)

ID	Story	Priority
PRD-US-DEV-001	As a developer, I want to generate and manage API keys, so that I can authenticate my application.	Must
PRD-US-DEV-002	As a developer, I want to access comprehensive API documentation, so that I can integrate with the platform.	Must
PRD-US-DEV-003	As a developer, I want to test API endpoints using an interactive console, so that I can validate my integration.	Should
PRD-US-DEV-004	As a developer, I want to download SDKs (Python, Node.js, Java), so that I can integrate quickly.	Should

5.21 Notifications (PRD-US-NOT)

ID	Story	Priority
PRD-US-NOT-001	As a user, I want to receive a notification when a document finishes indexing, so that I know it's ready.	Should
PRD-US-NOT-002	As an admin, I want to receive notifications when new users join my organization, so that I can assign roles.	Should

PRD-US-NOT-003	As a user, I want to configure notification channels (email, in-app, Slack), so that I receive alerts in my preferred way.	Could

5.22 Integrations (PRD-US-INT)

ID	Story	Priority
PRD-US-INT-001	As an admin, I want to connect the platform to SharePoint, so that documents are automatically synced.	Should
PRD-US-INT-002	As an admin, I want to connect the platform to Confluence, so that documents are automatically synced.	Should
PRD-US-INT-003	As an admin, I want to connect the platform to Google Drive, so that documents are automatically synced.	Could
PRD-US-INT-004	As an admin, I want to connect the platform to Notion, so that documents are automatically synced.	Could
PRD-US-INT-005	As an admin, I want to configure sync schedules and incremental syncs, so that the knowledge base stays current without full re-indexing.	Should

5.23 RBAC (PRD-US-RBC)

ID	Story	Priority
PRD-US-RBC-001	As an admin, I want to create custom roles with granular permissions, so that I can define specific access patterns.	Must

PRD-US-RBC-002	As an admin, I want to assign roles to users or groups, so that they get the correct permissions.	Must
PRD-US-RBC-003	As an admin, I want to define a matrix of permissions per role (upload, query, admin, export), so that access is precise.	Must
PRD-US-RBC-004	As a security admin, I want to enforce the principle of least privilege, so that users only have the permissions they need.	Must

5.24 Compliance (PRD-US-CMP)

ID	Story	Priority
PRD-US-CMP-001	As an auditor, I want to ensure that all data for EU employees resides in the EU region, so that we comply with GDPR.	Must
PRD-US-CMP-002	As a compliance officer, I want to configure data retention and deletion policies, so that we comply with regulatory requirements.	Must
PRD-US-CMP-003	As a compliance officer, I want to generate compliance reports (SOC 2, ISO 27001), so that I can pass external audits.	Should

5.25 Security (PRD-US-SEC)

ID	Story	Priority
PRD-US-SEC-001	As an admin, I want to configure PII detection and redaction in uploaded	Must

	documents, so that sensitive data is protected.	
PRD-US-SEC-002	As an admin, I want to configure prompt injection protections, so that the system is secure from adversarial attacks.	Must
PRD-US-SEC-003	As an admin, I want to configure data loss prevention (DLP) policies, so that sensitive information cannot be exfiltrated.	Must
PRD-US-SEC-004	As a security admin, I want to receive alerts for anomalous access patterns, so that I can respond to potential breaches.	Should

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6. Functional Requirements

6.1 Authentication (PRD-FR-AUTH)

Purpose: Provide secure, enterprise-grade identity management with support for corporate identity providers.

Workflow: The user navigates to the platform login page and selects their corporate Identity Provider. The platform redirects to the IdP's authentication endpoint. Upon successful authentication, the IdP returns a signed token (JWT or SAML assertion) to the platform. The platform validates the token, extracts the user's identity, groups, and roles, and issues a platform session cookie.

Business Rules: Every request to the platform MUST include a valid session token. Token validation occurs before any data access. Session tokens MUST have a configurable expiration (default: 8 hours). Refresh tokens MUST be supported for seamless session extension.

Edge Cases: If the IdP is temporarily unavailable, the platform MUST display a clear error message and offer a retry option. If the user's groups have changed since the last login, the platform MUST re-evaluate permissions on the next request.

Failure Handling: If token validation fails, the platform MUST reject the request with a 401 status and redirect to the login page. If the IdP returns an error, the platform MUST log the error and display a user-friendly message.

Acceptance Criteria:

- The system SHALL support SAML 2.0, OIDC, and OAuth 2.0 authentication flows.
- The system SHALL validate authentication tokens on every request.
- The system SHALL support configurable session expiration.
- The system SHALL log all authentication events (success, failure, session timeout).

Priority: Must

Dependencies: Identity Provider integration

6.2 Organizations (PRD-FR-ORG)

Purpose: Enable multi-tenant organization management with complete data isolation.

Workflow: An organization owner creates an organization during signup. The platform provisions a unique tenant ID. The owner invites users, who are then associated with the tenant. All subsequent data operations are scoped to the tenant ID.

Business Rules: Every resource (workspace, document, conversation, audit log) belongs to exactly one organization. Cross-tenant data access is technically impossible by design.

Acceptance Criteria:

- The system SHALL provision a unique tenant ID for each organization.
- The system SHALL enforce tenant isolation at the database and application layers.
- The system SHALL prevent any cross-tenant data access.

Priority: Must

6.3 Users (PRD-FR-USR)

Purpose: Manage user accounts within organizations.

Workflow: Users are created when invited by an organization owner or provisioned via SCIM. Each user has an identity, associated roles, and group memberships. Users can be deactivated (soft delete) or permanently removed.

Business Rules: A user can belong to multiple organizations but only one organization at a time within a session. Deactivated users retain their historical data (conversations, audit logs) for compliance purposes.

Acceptance Criteria:

- The system SHALL support user creation via invitation and SCIM.
- The system SHALL support user deactivation and reactivation.

- The system SHALL preserve historical data for deactivated users for configurable retention periods.

Priority: Must

6.4 Roles (PRD-FR-ROL)

Purpose: Define and manage roles within organizations.

Workflow: Administrators create roles and assign granular permissions to each role. Roles are then assigned to users or groups.

Business Rules: The platform SHALL provide default roles (Owner, Admin, Editor, Viewer) with pre-configured permissions. Custom roles can extend or restrict default permissions.

Acceptance Criteria:

- The system SHALL provide at least four default roles (Owner, Admin, Editor, Viewer).
- The system SHALL allow creation of custom roles with granular permissions.
- The system SHALL enforce role-based access control for all data operations.

Priority: Must

6.5 Permissions (PRD-FR-PRM)

Purpose: Enforce fine-grained access control across the platform.

Workflow: Every user action is evaluated against the user's roles and permissions. If the user lacks the required permission, the action is denied and logged.

Business Rules: Permission checks occur before data access. The principle of least privilege MUST be enforced by default.

Acceptance Criteria:

- The system SHALL enforce permission checks before every data operation.
- The system SHALL log all permission denials.
- The system SHALL support attribute-based access control (ABAC) for document-level permissions.

Priority: Must

6.6 Workspaces (PRD-FR-WS)

Purpose: Provide logical separation of knowledge bases within an organization.

Workflow: Administrators create workspaces and assign users and permissions. Each workspace contains its own documents, collections, and knowledge bases.

Acceptance Criteria:

- The system SHALL support multiple workspaces per organization.
- The system SHALL enforce workspace-level access controls.
- The system SHALL provide workspace-level analytics.

Priority: Must

6.7 Projects (PRD-FR-PRJ)

Purpose: Enable team-based organization of knowledge base work.

Workflow: Within a workspace, administrators can create projects that group related knowledge bases, documents, and team members. Projects provide a higher level of organization for complex knowledge management scenarios.

Acceptance Criteria:

- The system SHALL support projects within workspaces.
- The system SHALL allow project-level access control.

Priority: Could

6.8 Knowledge Bases (PRD-FR-KB)

Purpose: Define the scope of documents that can be queried by the AI.

Workflow: Administrators create a knowledge base by selecting one or more collections or folders. They configure the embedding model, chunking strategy, and retrieval settings.

Acceptance Criteria:

- The system SHALL support creating knowledge bases from collections or folders.
- The system SHALL allow configuration of embedding models per knowledge base.
- The system SHALL support configurable chunking strategies (fixed-size, semantic, hierarchical).

Priority: Must

6.9 Collections (PRD-FR-COL)

Purpose: Group documents by topic or purpose for focused retrieval.

Workflow: Users or administrators create collections and add documents to them. Collections can be queried independently or as part of a knowledge base.

Acceptance Criteria:

- The system SHALL support creation and management of collections.

- The system SHALL enforce collection-level access controls.

Priority: Must

6.10 Folders (PRD-FR-FLD)

Purpose: Provide hierarchical organization of documents within collections.

Workflow: Users create folder structures to organize documents. Folders inherit permissions from their parent collection unless explicitly overridden.

Acceptance Criteria:

- The system SHALL support nested folder structures.
- The system SHALL support folder-level permission inheritance.

Priority: Should

6.11 Document Upload (PRD-FR-UPL)

Purpose: Ingest documents into the platform for indexing and retrieval.

Workflow: Users upload documents via the web interface or API. The platform validates the file, scans for malware, extracts text, generates embeddings, and indexes the chunks.

Business Rules: The system SHALL support the following formats: PDF, DOCX, PPTX, XLSX, CSV, JSON, Markdown, HTML, TXT, and EPUB. Maximum file size per upload is 100 MB. Maximum total documents per knowledge base is 10 million.

Edge Cases: Password-protected PDFs require a decryption key or are skipped. Corrupted files are quarantined. Files with unsupported formats are rejected with a clear error message.

Failure Handling: Failed uploads are retried up to three times with exponential backoff. Persistent failures are quarantined and reported to the administrator.

Acceptance Criteria:

- The system SHALL support at least 12 document formats.
- The system SHALL quarantine files failing malware scans.
- The system SHALL retry failed uploads up to three times.

Priority: Must

Dependencies: Malware scanning service, text extraction engine, embedding model

6.12 Versioning (PRD-FR-VER)

Purpose: Track document versions and enable rollback.

Workflow: When a document is re-uploaded, the platform creates a new version. Previous versions are preserved and remain accessible for audit purposes.

Acceptance Criteria:

- The system SHALL maintain version history for all documents.
- The system SHALL allow rollback to any previous version.
- The system SHALL update the index when a new version is published.

Priority: Should

6.13 Metadata (PRD-FR-MET)

Purpose: Associate structured information with documents for filtering and discovery.

Workflow: During ingestion, the platform automatically extracts metadata (title, author, date, file type, page count). Users can add custom metadata fields and values.

Acceptance Criteria:

- The system SHALL extract standard metadata fields automatically.
- The system SHALL support custom metadata fields defined by administrators.
- The system SHALL support metadata-based filtering during search.

Priority: Must

6.14 OCR (PRD-FR-OCR)

Purpose: Extract text from scanned documents and images.

Workflow: When a PDF or image is uploaded, the platform detects whether the document contains embedded text or requires OCR. If OCR is needed, the document is processed through an OCR engine.

Acceptance Criteria:

- The system SHALL perform OCR on scanned documents automatically.
- The system SHALL support at least 50 languages for OCR.
- The system SHALL achieve > 95% OCR accuracy on clean documents.

Priority: Must

6.15 Image Extraction (PRD-FR-IMG)

Purpose: Extract and describe images within documents for retrieval.

Workflow: During document processing, images are extracted and processed through a vision model to generate descriptions. These descriptions are indexed alongside the surrounding text.

Acceptance Criteria:

- The system SHALL extract images from PDFs and Office documents.
- The system SHALL generate text descriptions for extracted images.
- The system SHALL index image descriptions for retrieval.

Priority: Should

6.16 PDF Support (PRD-FR-PDF)

Purpose: Comprehensive handling of PDF documents.

Workflow: PDFs are processed using a combination of native text extraction and OCR for scanned pages. Page numbers, headers, footers, and table structures are preserved.

Acceptance Criteria:

- The system SHALL extract text from both native and scanned PDFs.
- The system SHALL preserve page numbers and document structure.
- The system SHALL handle encrypted PDFs by prompting for a password.

Priority: Must

6.17 Office Documents (PRD-FR-OFF)

Purpose: Process Microsoft Office and Google Workspace documents.

Workflow: DOCX, PPTX, and XLSX files are processed using format-specific parsers that preserve document structure, styles, and metadata.

Acceptance Criteria:

- The system SHALL support DOCX, PPTX, and XLSX formats.
- The system SHALL preserve document structure (headings, lists, tables).
- The system SHALL extract metadata from Office document properties.

Priority: Must

6.18 CSV (PRD-FR-CSV)

Purpose: Process structured data in CSV format.

Workflow: CSV files are parsed into rows and columns. Each row is treated as a potential chunk, and column headers are included in the chunk context for semantic understanding.

Acceptance Criteria:

- The system SHALL parse CSV files with configurable delimiters.
- The system SHALL include column headers in chunk context.
- The system SHALL handle CSV files up to 10 million rows.

Priority: Should

6.19 JSON (PRD-FR-JSON)

Purpose: Process structured data in JSON format.

Workflow: JSON files are parsed and flattened. Key-value pairs are converted into natural language sentences for embedding generation.

Acceptance Criteria:

- The system SHALL parse nested JSON structures.
- The system SHALL convert key-value pairs into natural language for embedding.

Priority: Should

6.20 Markdown (PRD-FR-MKD)

Purpose: Process Markdown documents preserving structure.

Workflow: Markdown files are parsed, preserving headings, code blocks, lists, and links. Headings are used as natural chunk boundaries.

Acceptance Criteria:

- The system SHALL preserve Markdown structure during processing.
- The system SHALL use headings as chunk boundaries.

Priority: Must

6.21 HTML (PRD-FR-HTML)

Purpose: Process HTML documents and web pages.

Workflow: HTML content is cleaned of scripts and styles, and the remaining text content is extracted with structural context (headings, paragraphs, lists).

Acceptance Criteria:

- The system SHALL extract text content from HTML while removing scripts and styles.
- The system SHALL preserve structural context (headings, paragraphs).

Priority: Should

6.22 Audio Transcription (PRD-FR-AUD)

Purpose: Convert audio files to searchable text.

Workflow: Audio files (MP3, WAV, M4A) are transcribed using a speech-to-text model. The transcript is then processed as a regular text document for indexing.

Acceptance Criteria:

- The system SHALL transcribe audio files in at least 5 formats.
- The system SHALL support at least 10 languages for transcription.
- The system SHALL timestamp transcript segments for context retrieval.

Priority: Could

6.23 Video Transcription (PRD-FR-VID)

Purpose: Convert video files to searchable text.

Workflow: Video files (MP4, WebM, AVI) have their audio extracted and transcribed. Video frames may also be processed for visual content description.

Acceptance Criteria:

- The system SHALL transcribe audio from video files.
- The system SHALL timestamp transcript segments.

Priority: Could

6.24 Chunking (PRD-FR-CHK)

Purpose: Split documents into semantically meaningful segments for embedding.

Workflow: Documents are split into chunks based on the configured strategy. Supported strategies include fixed-size (with overlap), semantic (using sentence boundary detection), and hierarchical (preserving document structure).

Business Rules: Default chunk size is 512 tokens with 50-token overlap. Administrators can configure chunk size per knowledge base.

Acceptance Criteria:

- The system SHALL support at least three chunking strategies.
- The system SHALL preserve semantic coherence within chunks.
- The system SHALL support configurable chunk size and overlap.

Priority: Must

6.25 Embedding Generation (PRD-FR-EMB)

Purpose: Convert text chunks into dense vector representations.

Workflow: Each chunk is sent to an embedding model (e.g., OpenAI text-embedding-3-large, Cohere embed-v3) which returns a dense vector. The vector is stored alongside the chunk text and metadata in the vector database.

Acceptance Criteria:

- The system SHALL support at least three embedding model providers.
- The system SHALL batch embedding generation for efficiency.
- The system SHALL re-embed chunks when the embedding model is changed.

Priority: Must

6.26 Indexing (PRD-FR-IDX)

Purpose: Build and maintain the search index.

Workflow: Embedded chunks are indexed in the vector database with their associated metadata (tenant ID, ACL, document ID, chunk text, embedding vector). The index supports both vector similarity search and keyword search.

Acceptance Criteria:

- The system SHALL index all chunks within 60 seconds of embedding generation.
- The system SHALL support incremental indexing (no full re-index required).
- The system SHALL maintain index consistency across replicas.

Priority: Must

6.27 Vector Search (PRD-FR-VSR)

Purpose: Perform semantic similarity search across indexed chunks.

Workflow: The user query is embedded using the same model as the indexed chunks. The vector database returns the Top-K most similar chunks based on cosine similarity or another distance metric.

Acceptance Criteria:

- The system SHALL perform vector search within 200ms for indices up to 1 billion vectors.
- The system SHALL support cosine similarity and Euclidean distance metrics.

Priority: Must

6.28 Hybrid Search (PRD-FR-HSR)

Purpose: Combine vector search and keyword search for improved recall.

Workflow: The query is processed by both the vector search engine and a keyword search engine (e.g., BM25). Results are merged and reranked using a reciprocal rank fusion or learned model.

Acceptance Criteria:

- The system SHALL support hybrid search combining vector and keyword results.
- The system SHALL support configurable weighting between vector and keyword scores.

Priority: Must

6.29 Keyword Search (PRD-FR-KWS)

Purpose: Perform exact-match and fuzzy keyword search.

Workflow: The query is tokenized and matched against an inverted index of the document corpus. Results are ranked by relevance using BM25 or TF-IDF scoring.

Acceptance Criteria:

- The system SHALL support BM25-based keyword search.
- The system SHALL support fuzzy matching with configurable edit distance.

Priority: Must

6.30 Semantic Search (PRD-FR-SEM)

Purpose: Understand the meaning of the query beyond exact keyword matches.

Workflow: The query embedding is compared against chunk embeddings using cosine similarity. Results are ranked by semantic relevance.

Acceptance Criteria:

- The system SHALL perform semantic search with latency under 200ms.
- The system SHALL support multilingual semantic search.

Priority: Must

6.31 Metadata Filtering (PRD-FR-MTF)

Purpose: Narrow search results using structured metadata.

Workflow: Before or during the search, metadata filters (document type, date range, author, ACL) are applied to restrict the search space.

Acceptance Criteria:

- The system SHALL support filtering by at least 10 metadata fields.
- The system SHALL apply filters before retrieval to reduce search space.

Priority: Must

6.32 Reranking (PRD-FR-RRK)

Purpose: Improve search result quality by re-evaluating top results.

Workflow: The top-K results from the initial search are re-scored using a cross-encoder model that evaluates the relevance of each chunk to the query.

Acceptance Criteria:

- The system SHALL support cross-encoder reranking.
- The system SHALL improve retrieval precision by at least 15% compared to vector search alone.

Priority: Must

6.33 LLM Gateway (PRD-FR-LLM)

Purpose: Provide a unified interface for multiple LLM providers.

Workflow: The platform routes LLM requests through a gateway that supports multiple providers (OpenAI, Anthropic, Google, AWS Bedrock). The gateway handles rate limiting, retry logic, cost tracking, and provider selection.

Acceptance Criteria:

- The system SHALL support at least four LLM providers.
- The system SHALL support configurable provider selection per workspace.
- The system SHALL implement rate limiting and retry logic.

Priority: Must

6.34 Chat (PRD-FR-CHT)

Purpose: Provide a conversational interface for querying the knowledge base.

Workflow: The user types a question. The platform retrieves relevant context, constructs a prompt, sends it to the LLM, and streams the response back to the user.

Business Rules: The system prompt MUST instruct the LLM to only use the provided context and cite its sources. If the context is insufficient, the LLM MUST state that it cannot answer.

Acceptance Criteria:

- The system SHALL stream responses in real-time.
- The system SHALL provide citations for every factual claim.

- The system SHALL refuse to answer when context is insufficient.

Priority: Must

6.35 Streaming Responses (PRD-FR-STR)

Purpose: Deliver LLM responses incrementally for immediate user feedback.

Workflow: The LLM API response is streamed as server-sent events (SSE) or WebSocket. Each chunk of text is sent to the client as it is generated.

Acceptance Criteria:

- The system SHALL deliver the first token within 1 second of query submission.
- The system SHALL support graceful interruption of streaming responses.

Priority: Must

6.36 Conversation Memory (PRD-FR-MEM)

Purpose: Maintain context across multiple turns in a conversation.

Workflow: Previous conversation turns are included in the LLM prompt to provide context. The conversation history is truncated when it exceeds the LLM's context window, using summarization for older turns.

Acceptance Criteria:

- The system SHALL maintain conversation context for at least 20 turns.
- The system SHALL summarize older turns when the context window is exceeded.

Priority: Must

6.37 Prompt Templates (PRD-FR-TMP)

Purpose: Provide reusable prompt configurations.

Workflow: Administrators create prompt templates that define the system prompt, retrieval configuration, and output format. Users can select a template when starting a new conversation.

Acceptance Criteria:

- The system SHALL support creation and management of prompt templates.
- The system SHALL allow administrators to share templates organization-wide.

Priority: Should

6.38 Grounded Citations (PRD-FR-GCT)

Purpose: Ensure every AI-generated claim is traceable to a source document.

Workflow: The LLM is instructed to cite its sources using inline references. The platform validates that each citation corresponds to a retrieved document.

Acceptance Criteria:

- The system SHALL provide at least one citation per paragraph in the response.
- The system SHALL validate that citations reference actual retrieved documents.

Priority: Must

6.39 Feedback (PRD-FR-FDB)

Purpose: Collect user feedback to improve system performance.

Workflow: Users can provide thumbs up/down feedback on responses. Feedback is logged and used for future model evaluation and prompt engineering.

Acceptance Criteria:

- The system SHALL collect feedback on every response.
- The system SHALL make feedback data available for analytics.

Priority: Should

6.40 Search History (PRD-FR-SRH)

Purpose: Track and manage user search activity.

Workflow: All search queries and their results are logged in the user's search history. Users can view, search, and delete their history.

Acceptance Criteria:

- The system SHALL maintain search history for at least 90 days.
- The system SHALL allow users to search their own history.

Priority: Should

6.41 Bookmarks (PRD-FR-BKM)

Purpose: Allow users to save important documents or search results.

Workflow: Users can bookmark documents or specific search results. Bookmarks are organized by collection and are searchable.

Acceptance Criteria:

- The system SHALL support bookmarking documents and search results.

- The system SHALL support searching within bookmarks.

Priority: Could

6.42 Document Preview (PRD-FR-PRV)

Purpose: Allow users to preview documents without downloading them.

Workflow: The platform renders a preview of the document in the browser. For PDFs, this uses the browser's native PDF viewer. For Office documents, the platform converts to HTML for rendering.

Acceptance Criteria:

- The system SHALL preview at least PDF, DOCX, and Markdown formats.
- The system SHALL render previews within 2 seconds.

Priority: Should

6.43 Annotations (PRD-FR-ANN)

Purpose: Allow users to annotate documents and chat responses.

Workflow: Users can highlight text in document previews and add notes. Annotations are saved per user and can be shared with others.

Acceptance Criteria:

- The system SHALL support text highlighting and note-taking.
- The system SHALL support sharing annotations with specific users.

Priority: Could

6.44 Sharing (PRD-FR-SHR)

Purpose: Enable sharing of documents, conversations, and knowledge bases.

Workflow: Users can generate share links with configurable permissions (view, comment, edit). Shared resources are accessible to anyone with the link within the organization.

Acceptance Criteria:

- The system SHALL support shareable links with configurable permissions.
- The system SHALL enforce organization-level access controls on shared resources.

Priority: Should

6.45 Import/Export (PRD-FR-IMP)

Purpose: Allow bulk import and export of data.

Workflow: Administrators can import documents in bulk via CSV manifest or API. Data can be exported as JSON, CSV, or ZIP archives.

Acceptance Criteria:

- The system SHALL support bulk import of at least 10,000 documents.
- The system SHALL support export in JSON and CSV formats.

Priority: Should

6.46 Admin Dashboard (PRD-FR-ADB)

Purpose: Provide a comprehensive management interface for administrators.

Workflow: Administrators access a centralized dashboard showing system health, user activity, document statistics, and configuration options.

Acceptance Criteria:

- The system SHALL display system health metrics (uptime, latency, error rate).
- The system SHALL display user activity metrics (DAU, WAU, MAU).
- The system SHALL display document statistics (count, size, indexing status).

Priority: Must

6.47 System Settings (PRD-FR-STS)

Purpose: Configure platform-wide and workspace-level settings.

Workflow: Administrators configure settings including default LLM provider, embedding model, chunk size, and security policies through the admin interface or API.

Acceptance Criteria:

- The system SHALL support configuration via both UI and API.
- The system SHALL validate configuration changes before applying them.

Priority: Must

6.48 API Keys (PRD-FR-API)

Purpose: Manage programmatic access to the platform.

Workflow: Developers generate API keys from the developer portal. Each key has configurable permissions and rate limits. Keys can be rotated or revoked.

Acceptance Criteria:

- The system SHALL support generating, rotating, and revoking API keys.

- The system SHALL support configurable permissions per API key.
- The system SHALL log all API key usage.

Priority: Must

6.49 Rate Limits (PRD-FR-RAT)

Purpose: Prevent abuse and ensure fair resource allocation.

Workflow: Rate limits are enforced per user, API key, and organization. Limits are configurable by administrators. Exceeded requests receive a 429 status with a retry-after header.

Acceptance Criteria:

- The system SHALL enforce rate limits per user and API key.
- The system SHALL return 429 status with retry-after header for exceeded limits.
- The system SHALL support configurable rate limits per organization.

Priority: Must

6.50 Usage Dashboard (PRD-FR-USD)

Purpose: Provide visibility into platform usage and costs.

Workflow: The usage dashboard displays metrics including API calls, tokens consumed, storage used, and estimated costs. Data is broken down by workspace, user, and time period.

Acceptance Criteria:

- The system SHALL display usage metrics broken down by workspace and user.
- The system SHALL support configurable usage alerts.

Priority: Should

6.51 Analytics (PRD-FR-ANL)

Purpose: Provide insights into platform usage patterns and quality metrics.

Workflow: The analytics module collects and aggregates data on search queries, response quality, user engagement, and system performance.

Acceptance Criteria:

- The system SHALL provide search analytics (query volume, zero-result rate, average latency).

- The system SHALL provide response quality metrics (citation accuracy, hallucination rate).
- The system SHALL provide user engagement metrics (DAU, WAU, MAU, retention).

Priority: Should

6.52 Audit Logs (PRD-FR-ALG)

Purpose: Maintain an immutable record of all platform activities.

Workflow: Every significant action (authentication, data access, configuration change, query) generates an audit log entry. Logs are immutable and retained according to configured policies.

Acceptance Criteria:

- The system SHALL log all authentication, data access, and configuration events.
- The system SHALL make audit logs immutable (no modification or deletion).
- The system SHALL support configurable log retention periods.
- The system SHALL support export in CSV and JSON formats.

Priority: Must

6.53 Notifications (PRD-FR-NTF)

Purpose: Inform users and administrators of important events.

Workflow: The platform sends notifications via email, in-app alerts, and configurable webhooks. Users can configure their notification preferences.

Acceptance Criteria:

- The system SHALL support in-app, email, and webhook notifications.
- The system SHALL allow users to configure notification preferences.
- The system SHALL support notification templates.

Priority: Should

6.54 Webhooks (PRD-FR-WBH)

Purpose: Enable event-driven integrations with external systems.

Workflow: Administrators configure webhooks that are triggered by specific events (document indexed, user added, query completed). The platform sends a POST request with event data to the configured endpoint.

Acceptance Criteria:

- The system SHALL support configurable webhook endpoints.
- The system SHALL retry failed webhook deliveries with exponential backoff.
- The system SHALL support at least 10 event types for webhooks.

Priority: Should

6.55 Third-Party Integrations (PRD-FR-TPI)

Purpose: Connect the platform with external enterprise systems.

Workflow: The platform provides pre-built connectors for popular enterprise systems. Connectors use OAuth for authentication and sync data on a configurable schedule.

Acceptance Criteria:

- The system SHALL provide connectors for at least 8 enterprise systems.
- The system SHALL support incremental sync to minimize load on source systems.
- The system SHALL log all integration activities.

Priority: Should

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7. Non-Functional Requirements

7.1 Availability (PRD-NFR-AVL)

The platform MUST maintain 99.9% uptime (43.8 minutes of downtime per month maximum). No single point of failure shall exist in the critical path. Automatic failover within a region MUST be supported. Cross-region failover SHOULD be supported for disaster recovery scenarios. The platform MUST be capable of graceful degradation, maintaining core search and chat functionality even when non-critical components (analytics, notifications) are degraded.

Component	Target Availability	Maximum Downtime/Month
Authentication	99.99%	4.3 minutes
Search & Retrieval	99.95%	21.9 minutes
LLM Generation	99.9%	43.8 minutes
Document Ingestion	99.9%	43.8 minutes
Admin Dashboard	99.5%	3.6 hours

7.2 Scalability (PRD-NFR-SCL)

The platform MUST scale horizontally to support thousands of concurrent users and millions of indexed documents. The architecture MUST support:

- Up to 10,000 concurrent users per tenant.
- Up to 10 million indexed documents per tenant.
- Up to 1 billion indexed chunks per tenant.
- Up to 1,000 queries per second (QPS) per tenant.

Infrastructure MUST automatically scale based on load without manual intervention. The vector database MUST support horizontal scaling with consistent query latency.

7.3 Latency (PRD-NFR-LAT)

Operation	P50 Target	P95 Target	P99 Target
Authentication	< 50 ms	< 100 ms	< 200 ms
Document Upload	< 1 second	< 5 seconds	< 10 seconds
Vector Search	< 100 ms	< 200 ms	< 500 ms
Reranking	< 50 ms	< 100 ms	< 200 ms
LLM TTFB	< 500 ms	< 1 second	< 2 seconds
Full Response	< 5 seconds	< 10 seconds	< 15 seconds
Document Preview	< 1 second	< 2 seconds	< 5 seconds

7.4 Performance (PRD-NFR-PRF)

The platform MUST process document ingestion at a rate of at least 1,000 documents per hour per tenant. Embedding generation MUST support batching to minimize API costs and latency. The system MUST maintain consistent performance regardless of the total number of indexed documents within the stated capacity limits.

7.5 Reliability (PRD-NFR-REL)

No document loss SHALL occur during ingestion, indexing, or retrieval. The system MUST implement automatic retries with exponential backoff for transient failures. Indexing operations MUST be idempotent. Dead-letter queues MUST be used for failed ingestion jobs to enable manual review and retry.

7.6 Maintainability (PRD-NFR-MNT)

The platform MUST be designed for independent deployment of components. Configuration changes MUST not require code deployments. The system MUST support blue-green or canary deployments with zero-downtime releases.

7.7 Accessibility (PRD-NFR-ACC)

The platform MUST comply with WCAG 2.1 AA standards. All interactive elements MUST be keyboard-navigable. Screen reader compatibility MUST be verified. Color contrast MUST meet minimum ratios. The platform SHOULD support high-contrast and dark mode themes.

7.8 Localization (PRD-NFR-LCL)

The platform interface MUST support English as the default language. The platform SHOULD support at least 10 additional languages for the user interface. Document processing and search MUST support multilingual content.

7.9 Internationalization (PRD-NFR-I18N)

All user-facing strings MUST be externalized for translation. Date, time, and number formats MUST adapt to the user's locale. The platform MUST support right-to-left (RTL) languages.

7.10 Compliance (PRD-NFR-CMP)

The platform MUST comply with GDPR, SOC 2 Type II, ISO 27001, and HIPAA (for healthcare customers). Data residency requirements MUST be configurable per tenant. Audit trails MUST be immutable and retained for a minimum of 7 years.

7.11 Security (PRD-NFR-SCR)

All data MUST be encrypted at rest (AES-256) and in transit (TLS 1.2+). Secrets MUST be managed in a dedicated vault. API endpoints MUST be protected against common vulnerabilities (OWASP Top 10). The platform MUST undergo annual penetration testing.

7.12 Privacy (PRD-NFR-PRV)

The platform MUST support data subject access requests (DSAR). Users MUST be able to export and delete their personal data. PII detection MUST be performed during document ingestion. The platform MUST not retain user prompts or responses beyond the configured retention period.

7.13 Observability (PRD-NFR-OBS)

Every component MUST export structured metrics, logs, and distributed traces. The platform MUST provide a unified observability dashboard. Alerts MUST be configurable for key metrics (latency, error rate, throughput). Security events MUST be logged separately from operational logs.

7.14 Cost Efficiency (PRD-NFR-CST)

The platform MUST optimize for cost efficiency through batched embedding generation, prompt caching, and right-sized infrastructure. The system MUST provide cost visibility per tenant, workspace, and user. Auto-scaling MUST scale down during idle periods.

7.15 Disaster Recovery (PRD-NFR-DR)

The platform MUST support cross-region replication for disaster recovery. Recovery Point Objective (RPO) MUST be less than 1 hour. Recovery Time Objective (RTO) MUST be less than 4 hours. Automated backup and restore procedures MUST be tested quarterly.

7.16 Business Continuity (PRD-NFR-BCP)

The platform MUST maintain core functionality (search and chat) during partial outages. A documented Business Continuity Plan MUST be maintained. Incident response procedures MUST be tested semi-annually.

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8. Security Requirements

8.1 Enterprise Authentication (PRD-SEC-AUT)

The system MUST support Single Sign-On (SSO) via SAML 2.0, OpenID Connect (OIDC), and OAuth 2.0. Supported Identity Providers include Microsoft Entra ID (Azure AD), Okta, OneLogin, Google Workspace, and any standards-compliant IdP. Multi-Factor Authentication (MFA) MUST be enforced at the organization level or delegated to the Identity Provider. Session management MUST support configurable timeouts, concurrent session limits, and secure cookie attributes (HttpOnly, Secure, SameSite).

8.2 SSO (PRD-SEC-SSO)

The platform MUST support SAML 2.0 SP-initiated and IdP-initiated flows. The platform MUST support OIDC authorization code flow with PKCE. The platform MUST support Just-In-Time (JIT) user provisioning from the IdP. The platform MUST support automatic role assignment based on IdP groups.

8.3 OAuth (PRD-SEC-OAU)

The platform MUST support OAuth 2.0 for third-party application integration. The platform MUST implement the authorization code grant type with PKCE. Token scopes MUST be granular (read, write, admin per resource type).

8.4 OIDC (PRD-SEC-OID)

The platform MUST support OpenID Connect for identity federation. The platform MUST validate ID tokens and refresh tokens. The platform MUST support discovery endpoints (.well-known/openid-configuration).

8.5 SAML (PRD-SEC-SAM)

The platform MUST support SAML 2.0 for enterprise identity federation. The platform MUST validate SAML assertions and signatures. The platform MUST support SAML metadata exchange.

8.6 MFA (PRD-SEC-MFA)

The platform MUST support Time-based One-Time Passwords (TOTP), push notifications, and hardware security keys (WebAuthn/FIDO2). MFA enforcement MUST be configurable per organization and per user role.

8.7 RBAC (PRD-SEC-RBA)

The platform MUST implement Role-Based Access Control with at least four default roles (Owner, Admin, Editor, Viewer). The platform MUST support custom roles with granular permissions. Permission checks MUST be enforced at the API layer, not just the UI layer.

8.8 ABAC (PRD-SEC-ABA)

The platform MUST support Attribute-Based Access Control for document-level permissions. Access decisions MUST consider user attributes (department, clearance level, group membership), resource attributes (classification, sensitivity, owner), and environmental attributes (time, location, device).

8.9 Encryption (PRD-SEC-ENC)

All data MUST be encrypted at rest using AES-256-GCM. All data in transit MUST be encrypted using TLS 1.2 or higher. Embedding vectors MUST be encrypted in the vector database. The platform MUST support customer-managed encryption keys (CMEK) for data-at-rest encryption.

8.10 Secrets Management (PRD-SEC-SEC)

All secrets (API keys, database credentials, OAuth tokens) MUST be stored in a dedicated secrets management vault. Secrets MUST be rotated on a configurable schedule. Secrets MUST never be logged or exposed in error messages.

8.11 Audit Logging (PRD-SEC-AUD)

Every security-relevant event MUST generate an immutable audit log entry. Audit events include: authentication (success/failure), authorization decisions, data access, configuration changes, user provisioning/de-provisioning, and security policy changes. Audit logs MUST be tamper-evident and retained for a minimum of 7 years.

8.12 Secure Uploads (PRD-SEC-UPL)

Uploaded files MUST be scanned for malware using a dedicated scanning service. Files MUST be validated against their declared content type. The platform MUST reject files that do not match their declared type. Maximum file sizes MUST be enforced.

8.13 Malware Scanning (PRD-SEC-MLW)

All uploaded files MUST be scanned using a commercial malware detection engine (e.g., ClamAV, VirusTotal API). Files detected as malicious MUST be quarantined immediately. The administrator MUST be notified of quarantine events.

8.14 Data Isolation (PRD-SEC-DIS)

Data MUST be logically isolated between organizations using tenant IDs in all database queries. The platform MUST prevent SQL injection attacks through parameterized queries. The platform MUST prevent NoSQL injection attacks through input validation.

8.15 Tenant Isolation (PRD-SEC-TIS)

Every indexed chunk MUST include a tenant ID that is enforced in all queries. The vector database MUST enforce tenant isolation at the query level. The platform MUST prevent cross-tenant data access through both application-layer and database-layer controls.

8.16 PII Protection (PRD-SEC-PII)

The platform MUST detect PII (names, email addresses, phone numbers, SSNs, credit card numbers) in uploaded documents. PII detection MUST use a combination of regex patterns and NLP-based models. The platform MUST support configurable PII redaction policies (mask, hash, remove). PII detection MUST be performed before embedding generation.

8.17 Prompt Injection Mitigation (PRD-SEC-PIN)

The platform MUST implement multiple layers of prompt injection defense: input sanitization to remove or escape special characters, strict system prompt construction that treats retrieved context as untrusted, output validation to detect leaked instructions, and a content safety classifier to detect adversarial inputs.

8.18 Jailbreak Prevention (PRD-SEC-JBP)

The platform MUST detect and block jailbreak attempts in user queries. Detection methods MUST include pattern matching against known jailbreak templates and semantic classification of intent. Blocked queries MUST be logged and reported to the administrator.

8.19 Data Exfiltration Prevention (PRD-SEC-DEX)

The platform MUST prevent the LLM from generating content that could facilitate data exfiltration. This includes: refusing to generate SQL queries or API calls that could access external systems, refusing to encode sensitive data in alternative formats (base64, binary), and monitoring output for patterns consistent with data exfiltration attempts.

8.20 Secure APIs (PRD-SEC-AP)

All API endpoints MUST be protected by authentication and authorization. APIs MUST use HTTPS exclusively. APIs MUST implement input validation and output encoding. APIs MUST return appropriate error codes without exposing internal details. APIs MUST implement request size limits.

8.21 Rate Limiting (PRD-SEC-RTL)

The platform MUST enforce rate limits at multiple levels: per user (queries per minute), per API key (calls per second), and per organization (total throughput). Rate limiting MUST use a token bucket or sliding window algorithm. Exceeded requests MUST receive a 429 status with a retry-after header.

8.22 DDoS Mitigation (PRD-SEC-DDS)

The platform MUST be protected against Distributed Denial of Service attacks. Protection MUST include: CDN-level rate limiting, IP reputation-based filtering, and application-level request throttling. The platform MUST support integration with cloud provider DDoS protection services.

8.23 Compliance Considerations (PRD-SEC-CMP)

The platform MUST support GDPR data residency requirements by allowing customers to select their deployment region. The platform MUST support data deletion requests within

30 days. The platform MUST maintain a data processing agreement (DPA) with customers. The platform MUST support right to access and right to erasure requests.

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9. AI Features

9.1 Context Retrieval (PRD-AI-CRT)

The platform MUST retrieve relevant context from the knowledge base before generating a response. The retrieval pipeline MUST include: query preprocessing, candidate generation, relevance scoring, ACL filtering, reranking, and context assembly.

9.2 Query Rewriting (PRD-AI-QRW)

The platform SHOULD rewrite user queries to improve retrieval effectiveness. Query rewriting techniques MUST include: query expansion (adding synonyms), query decomposition (splitting complex queries), and hyphenation normalization.

9.3 Multi-Query Retrieval (PRD-AI-MQR)

The platform SHOULD generate multiple query variations from the original user query and perform parallel retrieval. The results from all variations are merged and deduplicated before reranking.

9.4 Hybrid Retrieval (PRD-AI-HRT)

The platform MUST support hybrid retrieval combining dense vector search with sparse keyword search (BM25). The platform SHOULD support configurable weighting between vector and keyword scores, with a default weight of 0.7 for vector and 0.3 for keyword.

9.5 Vector Retrieval (PRD-AI-VRT)

The platform MUST perform dense vector retrieval using cosine similarity or Euclidean distance. The platform SHOULD support at least three embedding model providers (OpenAI, Cohere, local models).

9.6 Keyword Retrieval (PRD-AI-KRT)

The platform MUST perform keyword retrieval using BM25 scoring against an inverted index. The platform SHOULD support fuzzy matching with configurable edit distance (default: 2).

9.7 Metadata Filtering (PRD-AI-MTF)

The platform MUST apply metadata filters (tenant ID, ACL, document type, date range) during retrieval to ensure only authorized and relevant documents are returned.

9.8 MMR Retrieval (PRD-AI-MMR)

The platform SHOULD support Maximum Marginal Relevance (MMR) retrieval to balance relevance and diversity in the retrieved results. The lambda parameter SHOULD be configurable (default: 0.5).

9.9 Cross-Encoder Reranking (PRD-AI-RRK)

The platform MUST support cross-encoder reranking of the top-K retrieved results. The reranker SHOULD be a high-precision model (e.g., Cohere rerank-v3, BGE-reranker) that evaluates the relevance of each chunk to the query.

9.10 Context Compression (PRD-AI-CCP)

The platform SHOULD compress retrieved context to fit within the LLM's context window. Compression techniques MUST include: truncation with priority on most relevant chunks, summarization of older conversation turns, and token-efficient prompt construction.

9.11 Conversation Memory (PRD-AI-MEM)

The platform MUST maintain conversation context across multiple turns. The conversation history MUST be managed to stay within the LLM's context window using summarization for older turns. The platform SHOULD support configurable memory window size.

9.12 Citation Validation (PRD-AI-CVT)

The platform MUST validate that every citation in the generated response corresponds to an actual retrieved document. Invalid citations MUST be flagged and corrected if possible.

9.13 Hallucination Reduction (PRD-AI-HLR)

The platform MUST employ techniques to minimize hallucinations: strict grounding constraints in the system prompt, retrieval confidence thresholds, citation requirement enforcement, and refusal to answer when context is insufficient.

9.14 Answer Verification (PRD-AI-AVR)

The platform SHOULD verify generated answers against the retrieved context using a secondary model. The verification step compares the generated answer with the source context and flags discrepancies.

9.15 Confidence Scoring (PRD-AI-CSC)

The platform SHOULD provide a confidence score for each generated response. The confidence score is based on: retrieval relevance scores, citation strength, and answer verification results.

9.16 Tool Calling (PRD-AI-TCL)

The platform SHOULD support LLM tool calling to enable the model to invoke external functions (e.g., search API, database query, calculator) during response generation.

9.17 Structured Outputs (PRD-AI-STR)

The platform SHOULD support structured output generation using JSON schemas. The LLM MUST be constrained to produce outputs that conform to the specified schema.

9.18 Agent Workflows (PRD-AI-AGW)

The platform SHOULD support multi-step agent workflows where the LLM can plan, execute, and iterate on tasks. Agent workflows MUST respect the same security and access controls as direct user queries.

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10. Search Features

10.1 Autocomplete (PRD-SRC-AC)

The search interface MUST provide autocomplete suggestions as the user types. Suggestions SHOULD include: document titles, frequently searched terms, and semantic matches.

10.2 Typo Tolerance (PRD-SRC-TY)

The search MUST tolerate common spelling mistakes and typos. The system SHOULD support fuzzy matching with an edit distance of up to 2 for short terms and 3 for longer terms.

10.3 Filters (PRD-SRC-FL)

The search interface MUST support filtering by: document type, date range, author, source system, classification level, and custom metadata fields. Filters MUST be combinable.

10.4 Advanced Search (PRD-SRC-AS)

The platform SHOULD provide an advanced search interface with field-specific search syntax, proximity operators, and negation.

10.5 Boolean Search (PRD-SRC-BS)

The platform MUST support Boolean search operators (AND, OR, NOT) with proper precedence. The system SHOULD support parenthetical grouping.

10.6 Saved Searches (PRD-SRC-SS)

Users SHOULD be able to save search queries with their filters and parameters for future reuse.

10.7 Search Suggestions (PRD-SRC-SJ)

The platform SHOULD provide search suggestions based on: popular queries, related queries, and query history.

10.8 Search Analytics (PRD-SRC-SA)

The platform SHOULD provide search analytics including: query volume, zero-result rate, average result count, average click-through rate, and query-to-conversation conversion rate.

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11. Chat Features

11.1 Streaming (PRD-CHT-STR)

The chat interface MUST stream LLM responses in real-time using Server-Sent Events (SSE) or WebSocket. The first token MUST arrive within 1 second.

11.2 Threading (PRD-CHT-THR)

The chat interface MUST support conversation threading, allowing users to branch conversations into separate threads for different topics.

11.3 Conversation History (PRD-CHT-HIS)

The platform MUST maintain a searchable history of all conversations. Conversations SHOULD be searchable by content, date, and knowledge base.

11.4 Conversation Export (PRD-CHT-EXP)

Users SHOULD be able to export conversations in Markdown, PDF, or JSON format.

11.5 Pinned Conversations (PRD-CHT-PIN)

Users SHOULD be able to pin important conversations to the top of their conversation list.

11.6 Shared Chats (PRD-CHT-SHR)

Users SHOULD be able to share conversations with specific colleagues, with configurable permissions (view-only or editable).

11.7 Chat Templates (PRD-CHT-TMP)

Administrators SHOULD be able to create chat templates that pre-configure the knowledge base, system prompt, and output format.

11.8 Regenerate (PRD-CHT-REG)

Users SHOULD be able to regenerate the last AI response with different parameters or after modifying the retrieved context.

11.9 Continue Generation (PRD-CHT-CON)

Users SHOULD be able to ask the AI to continue its response if it was cut off due to token limits.

11.10 Stop Generation (PRD-CHT-STP)

Users MUST be able to stop the AI's generation mid-stream.

11.11 Feedback (PRD-CHT-FDB)

Users SHOULD be able to provide feedback on responses (thumbs up/down) with optional comments.

11.12 Source Citations (PRD-CHT-CIT)

Every factual claim in the AI's response MUST include an inline citation linking to the source document.

11.13 Confidence Indicators (PRD-CHT-CID)

The chat interface SHOULD display confidence indicators for responses, such as color-coded badges (green = high confidence, yellow = medium, red = low).

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12. Administration

12.1 Tenant Management (PRD-ADM-TMT)

Administrators MUST be able to create, configure, and manage tenant settings including: organization name, deployment region, security policies, and billing configuration.

12.2 Workspace Management (PRD-ADM-WSM)

Administrators MUST be able to create, configure, archive, and delete workspaces. Workspace configuration includes: default LLM, embedding model, chunking strategy, and access policies.

12.3 Role Management (PRD-ADM-RLM)

Administrators MUST be able to create custom roles, assign permissions to roles, and manage role hierarchies.

12.4 User Provisioning (PRD-ADM-USP)

The platform MUST support user provisioning via: manual invitation, SCIM (System for Cross-domain Identity Management), and SSO-based JIT provisioning.

12.5 License Management (PRD-ADM-LIC)

Administrators MUST be able to manage license allocation, view license utilization, and purchase additional licenses.

12.6 Billing (PRD-ADM-BIL)

The platform MUST support subscription-based billing with usage-based components. Billing MUST be visible per tenant and per workspace.

12.7 Audit Logs (PRD-ADM-ALG)

Administrators MUST have access to a comprehensive, immutable audit log covering all platform activities. Logs MUST be searchable and exportable.

12.8 Analytics (PRD-ADM-ANL)

Administrators MUST have access to analytics dashboards showing: user adoption, search quality, response accuracy, and system performance.

12.9 System Health (PRD-ADM-HLT)

The platform MUST provide real-time system health monitoring including: component status, latency metrics, error rates, and resource utilization.

12.10 Feature Flags (PRD-ADM-FFL)

Administrators SHOULD be able to enable or disable features per organization or workspace using feature flags.

12.11 Configuration (PRD-ADM-CFG)

Administrators MUST be able to configure platform-wide settings including: default LLM provider, embedding model, chunk size, retention policies, and notification preferences.

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13. Integrations

13.1 Google Drive (PRD-INT-GD)

The platform SHOULD provide a connector for Google Drive that syncs documents (Docs, Sheets, Slides, PDFs) on a configurable schedule. Authentication MUST use Google OAuth 2.0.

13.2 Microsoft OneDrive (PRD-INT-OD)

The platform SHOULD provide a connector for Microsoft OneDrive that syncs documents on a configurable schedule. Authentication MUST use Microsoft Graph API with OAuth 2.0.

13.3 SharePoint (PRD-INT-SP)

The platform SHOULD provide a connector for SharePoint that syncs document libraries. The connector MUST support incremental sync and respect SharePoint permissions.

13.4 Dropbox (PRD-INT-DB)

The platform COULD provide a connector for Dropbox that syncs files and folders on a configurable schedule.

13.5 Confluence (PRD-INT-CF)

The platform SHOULD provide a connector for Atlassian Confluence that syncs spaces and pages. The connector MUST respect Confluence space permissions.

13.6 Notion (PRD-INT-NT)

The platform COULD provide a connector for Notion that syncs pages and databases using the Notion API.

13.7 Slack (PRD-INT-SL)

The platform SHOULD provide a connector for Slack that indexes public and private channel messages. The connector MUST respect Slack channel permissions.

13.8 Microsoft Teams (PRD-INT-TM)

The platform SHOULD provide a connector for Microsoft Teams that indexes channel messages and files.

13.9 GitHub (PRD-INT-GH)

The platform SHOULD provide a connector for GitHub that indexes repository content (markdown docs, README files, code comments).

13.10 GitLab (PRD-INT-GL)

The platform COULD provide a connector for GitLab with similar capabilities to the GitHub connector.

13.11 Jira (PRD-INT-JR)

The platform SHOULD provide a connector for Atlassian Jira that indexes issues, comments, and wiki pages.

13.12 REST API (PRD-INT-RA)

The platform MUST expose a comprehensive REST API for all core operations: document management, search, chat, user management, and configuration. The API MUST use JSON for request and response bodies.

13.13 Webhook Support (PRD-INT-WH)

The platform MUST support webhook-based event notifications for: document indexed, query completed, user action, and system events.

13.14 SDK Support (PRD-INT-SDK)

The platform MUST provide official SDKs for Python, Node.js, and Java. SDKs MUST include comprehensive documentation and examples.

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14. User Flows

14.1 Sign Up (PRD-FLW-SUP)

1. User navigates to the platform signup page.

2. User selects their corporate Identity Provider (or enters email for manual signup).
3. User authenticates with their IdP.
4. Platform creates an organization account and provisions a tenant ID.
5. User is assigned the Owner role.
6. User is redirected to the onboarding wizard.

14.2 Organization Creation (PRD-FLW-ORG)

1. Organization Owner accesses the organization settings.
2. Owner configures organization name, logo, and default settings.
3. Owner configures SSO/SCIM integration.
4. Owner invites initial administrators.
5. Organization is ready for use.

14.3 Workspace Creation (PRD-FLW-WSC)

1. Administrator navigates to the admin dashboard.
2. Administrator clicks "Create Workspace."
3. Administrator configures workspace name, description, and members.
4. Administrator configures default settings (LLM, embeddings, chunking).
5. Workspace is created and accessible to assigned members.

14.4 Upload Documents (PRD-FLW-UPL)

1. User navigates to the document upload interface.
2. User selects files (drag-and-drop or file picker).
3. Platform validates file types and sizes.
4. Platform scans files for malware.
5. Files are stored and queued for processing.
6. User sees upload progress indicators.

14.5 Index Documents (PRD-FLW-IDX)

1. Platform extracts text from uploaded documents (including OCR for scanned PDFs).
2. Platform generates metadata (title, author, date, page count).
3. Platform chunks the text based on configured strategy.

4. Platform generates embeddings for each chunk.
5. Platform indexes chunks in the vector database with metadata and ACLs.
6. Administrator receives a notification of completion.

14.6 Search (PRD-FLW-SRC)

1. User navigates to the search interface.
2. User enters a natural language query.
3. Platform rewrites the query (optional).
4. Platform performs hybrid search (vector + keyword) with ACL filtering.
5. Platform reranks results.
6. Results are displayed with previews, metadata, and relevance scores.

14.7 Chat (PRD-FLW-CHT)

1. User navigates to the chat interface.
2. User types a question.
3. Platform retrieves relevant context from the knowledge base.
4. Platform constructs the LLM prompt with system instructions, context, and user query.
5. LLM generates a response.
6. Response is streamed to the user with inline citations.

14.8 Citation Inspection (PRD-FLW-CIT)

1. User clicks on an inline citation in the chat response.
2. Platform displays the source document snippet.
3. Platform highlights the specific passage that supports the cited claim.
4. User can open the full document in the preview pane.

14.9 Admin Management (PRD-FLW-ADM)

1. Administrator navigates to the admin dashboard.
2. Administrator views system health, user activity, and document statistics.
3. Administrator modifies configuration settings.
4. Changes are applied and propagated across the platform.

14.10 User Invitation (PRD-FLW-INV)

1. Administrator navigates to user management.
2. Administrator enters user email addresses.
3. Administrator assigns roles and workspace access.
4. Platform sends invitation emails.
5. Users accept invitations and complete onboarding.

14.11 Permission Changes (PRD-FLW-PRM)

1. Administrator navigates to role management.
2. Administrator creates or modifies a role.
3. Administrator assigns the role to users or groups.
4. Permission changes are applied immediately.
5. Affected users see updated access on their next action.

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15. Acceptance Criteria

15.1 Authentication (PRD-AC-AUTH)

- AC-AUTH-001: 100% of authentication attempts are routed through the configured Identity Provider.
- AC-AUTH-002: Unauthorized access attempts are blocked with a 401 response and logged.
- AC-AUTH-003: MFA is enforced when configured at the organization level.
- AC-AUTH-004: Session tokens expire after the configured timeout period.

15.2 Document Ingestion (PRD-AC-DATA)

- AC-DATA-001: 100% of supported document formats are successfully ingested, chunked, and indexed.
- AC-DATA-002: Malware scanning detects and quarantines at least 99.9% of known malicious files.
- AC-DATA-003: OCR accuracy exceeds 95% for clean scanned documents.
- AC-DATA-004: Failed uploads are retried up to three times with exponential backoff.

15.3 Response Generation (PRD-AC-GEN)

- AC-GEN-001: 98% of generated responses include accurate, verifiable citations.
- AC-GEN-002: Hallucination rate is less than 2% for grounded queries.
- AC-GEN-003: The system refuses to answer when retrieved context is insufficient.

15.4 Performance (PRD-AC-PERF)

- AC-PERF-001: P95 search retrieval latency is under 200ms.
- AC-PERF-002: P95 LLM streaming TTFB is under 1 second.
- AC-PERF-003: P95 full response generation is under 10 seconds.
- AC-PERF-004: Document ingestion processes at least 1,000 documents per hour.

15.5 Security (PRD-AC-SEC)

- AC-SEC-001: The system detects and mitigates 100% of tested prompt injection payloads.
- AC-SEC-002: Cross-tenant data leakage is impossible (verified by automated penetration testing).
- AC-SEC-003: All data is encrypted at rest (AES-256) and in transit (TLS 1.2+).
- AC-SEC-004: PII detection identifies at least 95% of PII instances in test documents.

15.6 Availability (PRD-AC-AVL)

- AC-AVL-001: Platform maintains 99.9% uptime over a rolling 30-day window.
- AC-AVL-002: No single point of failure exists in the critical path.
- AC-AVL-003: Automatic failover occurs within 30 seconds of component failure.

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16. Success Metrics

Metric	Category	Target	Measurement
Weekly Active Users (WAU) / Total Users	Adoption	> 40%	Analytics pipeline
Monthly User Retention	Retention	> 85%	Analytics pipeline
P95 Search Latency	Latency	< 200ms	Observability

P95 Response TTFB	Latency	< 1 second	Observability
Search Relevance Score	Search Quality	> 4.5/5.0	User feedback
Grounding Accuracy	Grounding	> 98%	Citation validation
Citation Accuracy	Citation	> 95%	Automated evaluation
Hallucination Rate	Hallucination	< 2%	Automated evaluation
System Uptime	Uptime	99.9%	Monitoring
Indexing Throughput	Throughput	> 1,000 docs/hour	Operational metrics
User Satisfaction (NPS)	Satisfaction	> 50	Surveys
Time-to-First-Value	Adoption	< 7 days	Analytics

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17. Risks

17.1 Technical Risks

Risk	Likelihood	Impact	Mitigation
LLM hallucination despite grounding	Medium	High	Strict reranking, confidence thresholds, refusal to answer
Retrieval latency exceeds SLAs at scale	Medium	Medium	Optimized vector databases, caching, asynchronous indexing
Embedding model quality degradation	Low	High	Multi-model support, A/B testing, quality monitoring
LLM API provider outage	Medium	High	Multi-provider gateway, automatic failover
Vector database scaling bottleneck	Low	High	Horizontal scaling architecture, partitioning strategy

17.2 Business Risks

Risk	Likelihood	Impact	Mitigation
Low enterprise adoption	Medium	High	Onboarding wizard, success metrics, customer success team
Competing RAG platforms	High	Medium	Differentiation through security and compliance
LLM API pricing changes	Medium	Medium	Multi-provider strategy, cost optimization

17.3 Operational Risks

Risk	Likelihood	Impact	Mitigation
Data ingestion pipeline failures	Medium	Medium	Dead-letter queues, monitoring, retry logic
Insufficient operational staffing	Low	High	Automation, managed services, runbooks
Customer support overload	Medium	Medium	Self-service documentation, in-app help

17.4 Security Risks

Risk	Likelihood	Impact	Mitigation
Data leakage across tenants	Low	Critical	Multi-layer isolation, penetration testing, audit logging
Prompt injection attacks	Medium	High	Defense-in-depth, input sanitization, output validation
Credential compromise	Low	Critical	Vault-based secrets management, rotation

			policies
Insider threat	Low	High	RBAC, audit logging, anomaly detection

17.5 AI Risks

Risk	Likelihood	Impact	Mitigation
Model output bias	Medium	Medium	Bias testing, output validation
Model capability regression	Low	High	Version pinning, regression testing
Context window overflow	Medium	Medium	Context compression, summarization

17.6 Compliance Risks

Risk	Likelihood	Impact	Mitigation
GDPR non-compliance	Low	Critical	Data residency controls, DSAR support, DPA
Regulatory changes	Medium	Medium	Compliance monitoring, flexible architecture
Audit failure	Low	High	Continuous compliance monitoring, automated reporting

17.7 Scalability Risks

Risk	Likelihood	Impact	Mitigation
Index size exceeds capacity	Low	High	Partitioning, sharding, capacity planning

Concurrent user load spike	Medium	Medium	Auto-scaling, load testing, rate limiting
Storage cost growth	Medium	Medium	Tiered storage, compression, archival policies

18. Assumptions

1. Enterprise customers will provide and manage their own Identity Providers (IdPs) for authentication.
2. The volume of indexed documents per tenant will not exceed 10 million initially (expandable through scaling).
3. Customers are responsible for configuring and maintaining their own document repositories (e.g., SharePoint, Confluence).
4. LLM APIs provided by third-party vendors (e.g., OpenAI, Anthropic, Google) will remain available and performant.
5. Customers have internet connectivity to the cloud platform.
6. Customers will comply with the platform's terms of service and acceptable use policy.
7. The platform's cloud infrastructure provider (e.g., AWS, GCP, Azure) will maintain their SLAs.
8. Embedding models will continue to improve in quality over time.
9. The platform will be deployed in a single cloud region per tenant (multi-region is a future capability).
10. Customers will allocate at least one dedicated administrator for platform management.

19. Constraints

19.1 Budget

Initial deployment must optimize for cloud-native managed services to minimize operational overhead. The platform must achieve profitability within 18 months of launch. Infrastructure costs per tenant must decrease over time through economies of scale.

19.2 Infrastructure

The platform MUST support multi-region deployment to comply with data residency laws. The platform MUST be deployed on a major cloud provider (AWS, GCP, or Azure). On-premise deployment is not supported in the initial release.

19.3 Compliance

The system MUST adhere to SOC 2 Type II and ISO 27001 standards. GDPR compliance is mandatory for EU customers. HIPAA compliance is required for healthcare customers. The platform must support data deletion within 30 days of request.

19.4 Technology

The architecture must remain implementation-independent, avoiding lock-in to a single vector database or LLM provider. All components MUST be replaceable without affecting the platform's core functionality. The platform MUST use containerized deployment (Kubernetes or equivalent).

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20. Future Roadmap

20.1 Phase 1: Core Foundation (Months 1-3)

Phase 1 establishes the foundational platform capabilities. This phase includes multi-tenant architecture setup with tenant isolation, basic document ingestion supporting PDF and Markdown formats, vector search with a single embedding model, simple LLM generation with a single provider, SSO authentication with a single IdP, and a basic chat interface with citations.

20.2 Phase 2: Enterprise Enhancements (Months 4-6)

Phase 2 adds enterprise-grade capabilities. This phase includes advanced hybrid search with BM25 and vector combination, cross-encoder reranking, comprehensive RBAC and ACL enforcement, integration connectors for SharePoint and Confluence, audit logging with immutable records, compliance reporting, and an admin dashboard with analytics.

20.3 Phase 3: AI Agents and Extensibility (Months 7-9)

Phase 3 introduces advanced AI capabilities and developer extensibility. This phase includes agent workflows with tool calling, advanced query rewriting and context compression, public REST API and SDK release (Python, Node.js, Java), custom prompt templates and shared chats, webhook-based event notifications, and multi-modal RAG (image and audio support).

20.4 Enterprise Roadmap

The enterprise roadmap extends beyond Phase 3 with on-premise deployment options for air-gapped environments, advanced SCIM provisioning with automatic role assignment, custom branding and white-label options, dedicated instance deployment for enterprise customers, and advanced compliance features (FedRAMP, SOC 2 Type II certification).

20.5 AI Roadmap

The AI roadmap focuses on advancing the platform's intelligence. Planned capabilities include fine-tuning capabilities for domain-specific models, multi-modal RAG supporting video and image content, autonomous agentic actions with tool use, advanced hallucination detection and correction, and adaptive retrieval strategies that learn from user feedback.

Appendix A: Glossary

Term	Definition
RAG	Retrieval-Augmented Generation — a technique that augments LLM responses with retrieved external knowledge
LLM	Large Language Model — a neural network trained on large text corpora
Embedding	A dense numerical vector representation of text
Chunk	A segment of a document used for embedding and retrieval
Vector Database	A database optimized for storing and querying dense vector embeddings
Tenant	An organization that uses the platform
ACL	Access Control List — a set of permissions attached to a resource
RBAC	Role-Based Access Control
ABAC	Attribute-Based Access Control
SSO	Single Sign-On
OIDC	OpenID Connect

SAML	Security Assertion Markup Language
MFA	Multi-Factor Authentication
SCIM	System for Cross-domain Identity Management
BM25	Best Matching 25 — a probabilistic retrieval function
MMR	Maximum Marginal Relevance
TTFB	Time to First Byte
SLA	Service Level Agreement
RPO	Recovery Point Objective
RTO	Recovery Time Objective
DLP	Data Loss Prevention
PII	Personally Identifiable Information
DSAR	Data Subject Access Request
DPA	Data Processing Agreement
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act

Appendix B: Requirement Traceability Matrix

Module	Requirements Count	Priority Distribution	Key IDs
Authentication	5	4 Must, 1 Should	PRD-US-AUTH-001 to 005
Organization	5	3 Must, 1 Should, 1 Could	PRD-US-ORG-001 to 005
Workspace	5	3 Must, 1 Should, 1 Could	PRD-US-WS-001 to 005
Document Management	9	4 Must, 4 Should, 1 Could	PRD-US-DOC-001 to 009

Folder Management	3	0 Must, 2 Should, 1 Could	PRD-US-FLD-001 to 003
Collections	3	1 Must, 1 Should, 1 Could	PRD-US-COL-001 to 003
Knowledge Bases	4	1 Must, 1 Should, 2 Could	PRD-US-KB-001 to 004
Search	5	1 Must, 3 Should, 1 Could	PRD-US-SRC-001 to 005
Chat	6	3 Must, 3 Should	PRD-US-CHT-001 to 006
Conversation History	4	1 Must, 1 Should, 2 Could	PRD-US-HIS-001 to 004
Citations	4	2 Must, 1 Should, 1 Could	PRD-US-CIT-001 to 004
Prompt Library	3	0 Must, 0 Should, 3 Could	PRD-US-PRO-001 to 003
Agents	3	0 Must, 0 Should, 3 Won't	PRD-US-AGT-001 to 003
Administration	5	2 Must, 1 Should, 2 Could	PRD-US-ADM-001 to 005
Billing	4	2 Must, 1 Should, 1 Could	PRD-US-BIL-001 to 004
Audit Logs	4	2 Must, 2 Should	PRD-US-AUD-001 to 004
Analytics	4	0 Must, 3 Should, 1 Could	PRD-US-ANA-001 to 004
Settings	3	0 Must, 2 Should, 1 Could	PRD-US-SET-001 to 003
API Usage	2	0 Must, 1 Should, 1 Could	PRD-US-API-001 to 002
Developer Portal	4	2 Must, 2 Should	PRD-US-DEV-001 to 004
Notifications	3	0 Must, 2 Should, 1 Could	PRD-US-NOT-001 to 003

Integrations	5	0 Must, 3 Should, 2 Could	PRD-US-INT-001 to 005
RBAC	4	3 Must, 1 Must	PRD-US-RBC-001 to 004
Compliance	3	2 Must, 1 Should	PRD-US-CMP-001 to 003
Security	4	3 Must, 1 Should	PRD-US-SEC-001 to 004

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